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B.Tech. Degree I Semester Examination November 2019

ME/SE 19-200-0103 A ENGINEERING GRAPHICS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(1 × 12 = 12)

- I. Draw an ellipse having major axis of 75 mm and minor axis of 50 mm. Also draw a tangent and normal to the curve from any point 'p' on the curve. (12)

OR

- II. Construct a diagonal scale with RF 1/5000 showing meters and long enough to measure upto I kilometer. Mark a distance of 648 meters on it. (12)

PART B

(4 × 15 = 60)

- III. (a) A line AB 70 mm long is inclined 30° to HP and 40° to VP. The point A is 10mm above HP and 20 mm in front of VP. Draw the projections of the line. Mark the traces. (9)

- (b) A hexagonal lamina of side 35 mm is resting on HP on one of its sides. The surface of the lamina is inclined 35° to HP. Draw the projections of the lamina. (6)

OR

- IV. (a) One end of a line AB is 10 mm above HP and 15 mm in front of VP. The other end of the line is 60 mm above HP and 50 mm in front of VP. Draw the projections of the line AB and find the true length of the line if the end projectors are 55 mm apart. (9)

- (b) A circular lamina of 50 mm diameter is resting on HP. The surface of the lamina is inclined 35° to HP. Draw the projections of the lamina. (6)

- V. A square pyramid of based edge 30mm and 60 mm height is resting on HP on one of its base edge. The axis of the pyramid is inclined 30° with HP. Base edge on which it rests makes an angle 40° with VP. Draw the final front view and top view. (15)

OR

- VI. A cone of base diameter 50 mm and height 60 mm is resting on HP. A cutting plane inclined 30° to HP and passing through the centre of the axis, cuts the cone. Draw the front view, sectional top view and true shape of the section. (15)

- VII. A square prism, edge of base 30 mm and height 60 mm, resting on its base on HP, is completely penetrated by another square prism of 20 mm base edge such that the axis of the penetrating prism is perpendicular to and 5 mm in front of the axis of the vertical prism. The rectangular faces of the two prisms are equally inclined to the VP. Draw the projections of the solids showing line of intersection. Assume any suitable length for penetrating prism. (15)

OR

(P.T.O.)

- VIII. A cone of base diameter 40 mm and height 60 mm is resting on HP on its base. An ant starts from one point on the base and reaches a point on the base which is diametrically opposite to the starting point through shortest path. Trace the path of the ant using radial development method. (15)
- IX. Draw the isometric view of a pentagonal pyramid side of base 40 mm and height 80 mm which rests centrally on a cylinder of diameter 120 mm and height 40 mm. (15)
- OR**
- X. Draw the perspective projection of a pentagonal prism of side 25 mm and length 50 mm, lying on one of its rectangular faces on the ground plane and one pentagonal face touching the picture plane. The station point is 55 mm in front of the picture plane and lies in the centre plane which is 75 mm to the left of the center of the prism. Station point is 30 mm above the ground plane. (15)
